

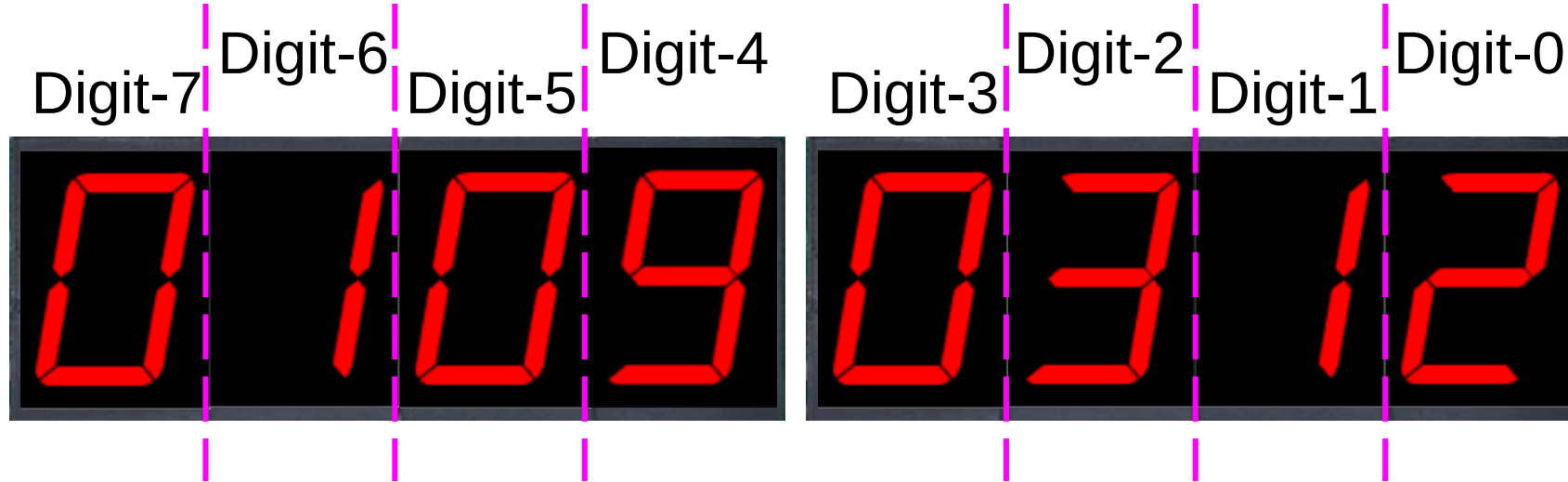
Project 1

Continuously display the student ID of a group member, leading with a “0”, e.g., **01090312**, on the 8-digit 7-segment LED display **at the specified order**.

- ✓ Let **the right-most digit** of 7-segment display be **the digit-0**, and **the left-most digit** one **the digit-7**.
- ✓ After the board is powered on, the 7-segment displays will take turn displaying their contents.
 - The order is: **digit-0 $\Rightarrow$ digit-2 $\Rightarrow$ digit-4 $\Rightarrow$ digit-6 $\Rightarrow$ digit-7 $\Rightarrow$ digit-5 $\Rightarrow$ digit-3 $\Rightarrow$ digit-1 $\Rightarrow$ digit-0 (repeat)**.

As for the time duration

- whenever a digit is on, it lasts 1.0 second.
- When all the digits are off, it lasts 0.5 second.



t = 0
(Unit: sec.)



t = 0.0 ~ 1.0



t = 1.0 ~ 1.5



t = 1.5 ~ 2.5



t = 2.5 ~ 3.0



t = 3.0 ~ 4.0



t = 4.0 ~ 4.5



t = 4.5 ~ 5.5



t = 5.5 ~ 6.0



t = 6.0 ~ 7.0

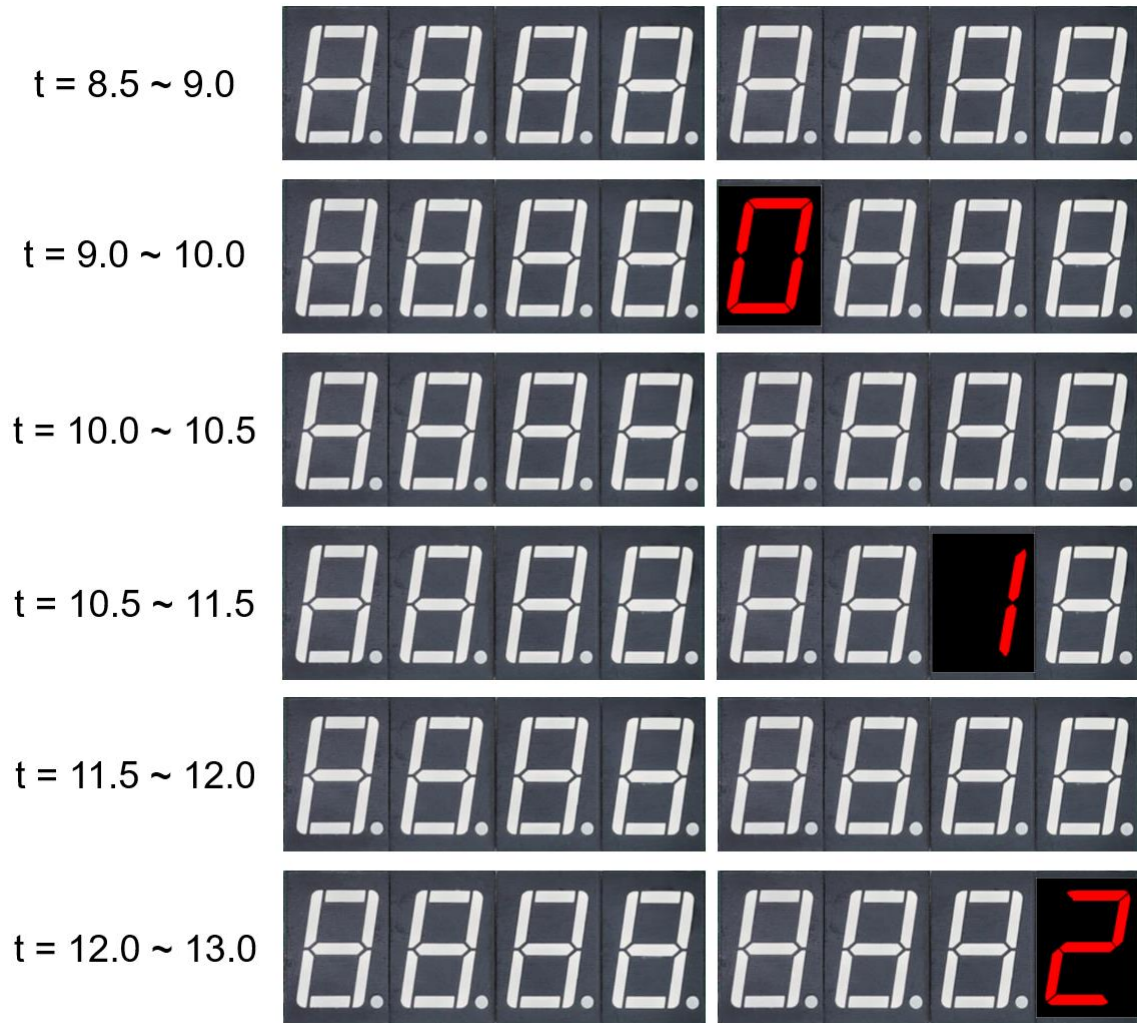


t = 7.0 ~ 7.5



t = 7.5 ~ 8.5





Repeat as the 2nd display in the last slide.

Materials to Submit

1. Source code in .asm file format.
2. A short video demonstration (no longer than 30 second).
3. A report which includes:
 - 1) Flow chart.
 - 2) Few pictures of 8051 board with intermediate results.
 - 3) Discussion of difficulties encountered and how your group resolve them .
 - 4) Acknowledge the help received, if any.

DO NOT put segments of source code in your report, unless you have something special and would like to show off.

Points to Note

1. In case get stuck, you are welcomed and encouraged to discuss with your friends/classmates.
2. Bonus will be awarded to the groups that provide help. It is your responsibility to acknowledge the help received, if any, in your report. Give a brief description on the difficulty you have and the help you receive.
3. Don't cheat. Don't copy any result (code or report) of other group.
Once caught, all groups involved will be failed for this project.
4. To avoid your work being stolen, don't give your report to other group.

Grading Policy

1. 60%: successful implementation of the program.
2. 30%: power point presentation.
3. 10%: style of program: succinct and structural program, comments are elegant and clear of information.